EnergyMind Neighborhood

Preliminary Design: Confidence Scores and Explainability

1 Goal 2: Confidence Scores and Explainability

1.1 Goal and Requirements

The challenge requires that every flagged event comes with a verdict that a human can understand and verify. The system must output a confidence score, provide a plain-language explanation, and maintain an auditable decision log that records what was detected, concluded and why. It should use interpretable approaches and avoid black-box scoring whose reasoning cannot be examined.

Anomalies in this project are equipment and measurement problems, suspected data manipulation (theft or false data injection) and genuine electrical events. The component therefore chooses among four verdicts, the same four that the escalation layer of Goal 3 routes: `NORMAL`, `GENUINE_EVENT` (the data is right and the grid did something), `METER_FAULT` (a faulty or corrupted meter) and `SUSPICIOUS_DATA` (readings that look manipulated). A fifth outcome, `NO_DATA`, records a communication loss, which is not scored. The prototype runs on node N1 of the EnergyMind dataset: one transformer, 30 smart meters on three feeders of 10, readings every 30 minutes. Table 1 maps each requirement to a design element of this section.

Requirement	**Design element**	**Artifact produced**
Confidence score	Calibrated probability of each of the four verdicts, obtained by multinomial Platt scaling of a few inspectable inputs (Section 1.6)	Fields `class_probabilities`, `confidence`, `band`; calibration curve, Brier score
Plain-language explanation	Rules + SHAP → evidence → structured summary → LLM text (no retrieval), checked against the rule trace, the SHAP ranking and the logged evidence before it is shown (Section 1.7)	Verdict card; field `llm_diagnosis` with the result of the faithfulness check
Auditable decision log	One hash-chained record per flagged meter-day (Section 1.8)	`decision_log.jsonl`
Record what was detected, concluded and why	Three explicit blocks in every record: `detected`, `concluded`, `why`	Rule trace, SHAP evidence, neighborhood support
Interpretable approaches	Rule-backed verdicts; SHAP on the detector scores; a calibration model whose coefficients can be printed	No verdict whose reasoning cannot be inspected

Table 1: Requirements of Challenge Goal 2 and where this design meets them.

1.2 Inputs and Outputs of this Component

The detection layer of Goal 1 works on *meter-days* (48 readings of one meter) and, for every flagged meter-day, passes on: the stack decision and each detector's score (rules, robust z, CUSUM, Isolation Forest, One-Class SVM, autoencoder); which rules fired, with the measured value and threshold; the neighborhood support (share of the other 9 meters of the feeder flagged the same day); the state of the feeder energy-balance alarm; and the autoencoder's per-feature reconstruction error. This component adds a few evidence quantities computed from the same day's readings (Section 1.4) and produces, for every flagged meter-day, a verdict with its sub-type, the four class probabilities and a confidence band, the supporting evidence, a plain-language explanation and one decision-log record. The class probabilities are the input of the Goal 3 escalation layer.

1.3 Design Principles

1. **The LLM only describes.** The language model never detects an anomaly and never changes a verdict or a confidence value. This follows the conclusion of the rule-aware prompting work that the main value of an LLM here is a readable rationale for the operator rather than a replacement for the detector [1].
2. **Every statement is traceable.** Each statement in an explanation points to a logged item: a rule that fired, a feature value, or a SHAP contribution.
3. **Confidence is measured.** The score is calibrated on labeled events and its quality is evaluated like any other output of the system (Section 1.9).
4. **No explanation without evidence.** If the generated text fails the faithfulness check, the structured summary is shown instead.

1.4 Evidence Available to the Explanation Layer

We index a flagged meter-day by meter i and day t. Goal 1 supplies the detector scores, the fired rules and the neighborhood support $s_{i,t}$; the neighborhood-support idea follows prior work that adjusts an anomaly score using temporal context and the neighborhood [7]. This component computes the evidence vector $\mathbf{e}_{i,t}$ of Table 2 from the meter's 48 readings. We write E for the energy imported in a 30-minute interval ($\Delta t = 0.5$ h), V, I and PF for voltage, current and power factor, V^{head} for the voltage measured at the head of the feeder and X for the exported PV energy. Baselines (profiles, medians, scales) are learned on clean meter-days only, i.e. days whose 48 readings are all labelled normal.

Symbol	Definition	What it tells a human
$v^0_{i,t}$	Share of readings with $V = 0$, or a `power_fail` entry in the meter's event log	Supply was lost: the meter is reporting a real outage
$g_{i,t}$	Share of missing readings	Communication loss
$\phi_{i,t}$	Longest run of readings in which energy *and* voltage are both unchanged (only while $V > 0$ and $X = 0$)	A stuck meter freezes every channel; real voltage never repeats for hours
$\rho_{i,t}$	Median over the day of $E \,/\, (V I \,\mathrm{PF}\, \Delta t/1000)$, using only readings with $V > 0$, $X = 0$ and an expected energy above 0.02 kWh	Electrical consistency: 1 when energy, voltage and current agree; skipped while the home exports PV
$s_{i,t}$	Share of the other 9 meters of the feeder flagged the same day (from Goal 1)	One house, part of the feeder, or the whole feeder
$\Delta V_{i,t}$	24-h median of $V - V^{\mathrm{head}}$	A meter cannot read more than its feeder head: a positive excess is a voltage-sensor offset
$\delta_{i,t}$	Relative drop of the day's energy below the meter's own 28-day profile	How much less than usual this home reports
$b_{i,t}$	Feeder energy-balance alarm (from Goal 1): on when the feeder's daily loss has moved away from its seasonal reference	Energy is disappearing (or appearing) somewhere on this feeder
$\kappa^{\mathrm{tamp}}_{i,t}$	`magnetic_tamper` or `cover_open` in the meter's event log	Weak hint of physical tampering (the log also has false alarms)
$c^{\mathrm{drift}}_{i,t}$	CUSUM alarm of Goal 1 on the meter's voltage offset, or slope of the meter's share of the feeder over 14 days	Slow drift or a persistent offset
$\pi_{i,t}$	Number of consecutive flagged days for this meter while its neighbors are not flagged	How long the deviation has lasted

Table 2: Evidence vector $\mathbf{e}_{i,t}$ of a flagged meter-day.

Why these quantities, on this dataset. Three facts of node N1 shaped the list. (i) A supply loss is recorded as zeros, not as missing data: in N1 the 683 supply-loss readings of outages, load shedding, cable faults and the transformer trip have $V = 0$, while only communication losses are blank. Without v^0 a flat zero would look like a frozen meter. (ii) The consistency ρ is meaningful because the dataset's *data-only* injections edit the energy field alone ($\rho \approx 0.65$–0.69, or 1.68 for inflated readings), while honest meters already carry about 1% measurement error. Without the export gate, ρ would fire on 39% of the clean readings of N1's three PV homes; with it, on 0.25% of all clean readings. (iii) Sensor bias and physical bypass scale energy *and* current together, so ρ stays at 1 for them; what reveals a bypass is a large drop δ against the meter's own history together with the feeder balance b.

1.5 Rule-Backed Verdict Layer

The rule-backed scoring turns the evidence vector into a verdict and a sub-type (Figure 1). Rules are evaluated in a fixed order, so the first matching rule is the one reported, and its identifier appears in the log and in the explanation. The order follows what makes the other evidence unreliable: a lost supply first (otherwise a flat zero looks like a frozen meter), then missing and frozen data, then physical consistency, which separates a manipulated or corrupted value from a real change, and only then the neighborhood support, which decides between a feeder-wide event and a local one. The split between sudden problems (R2, R3) and slow ones (R5, R5b) mirrors the split between point and contextual anomalies reported for smart-meter data [6].

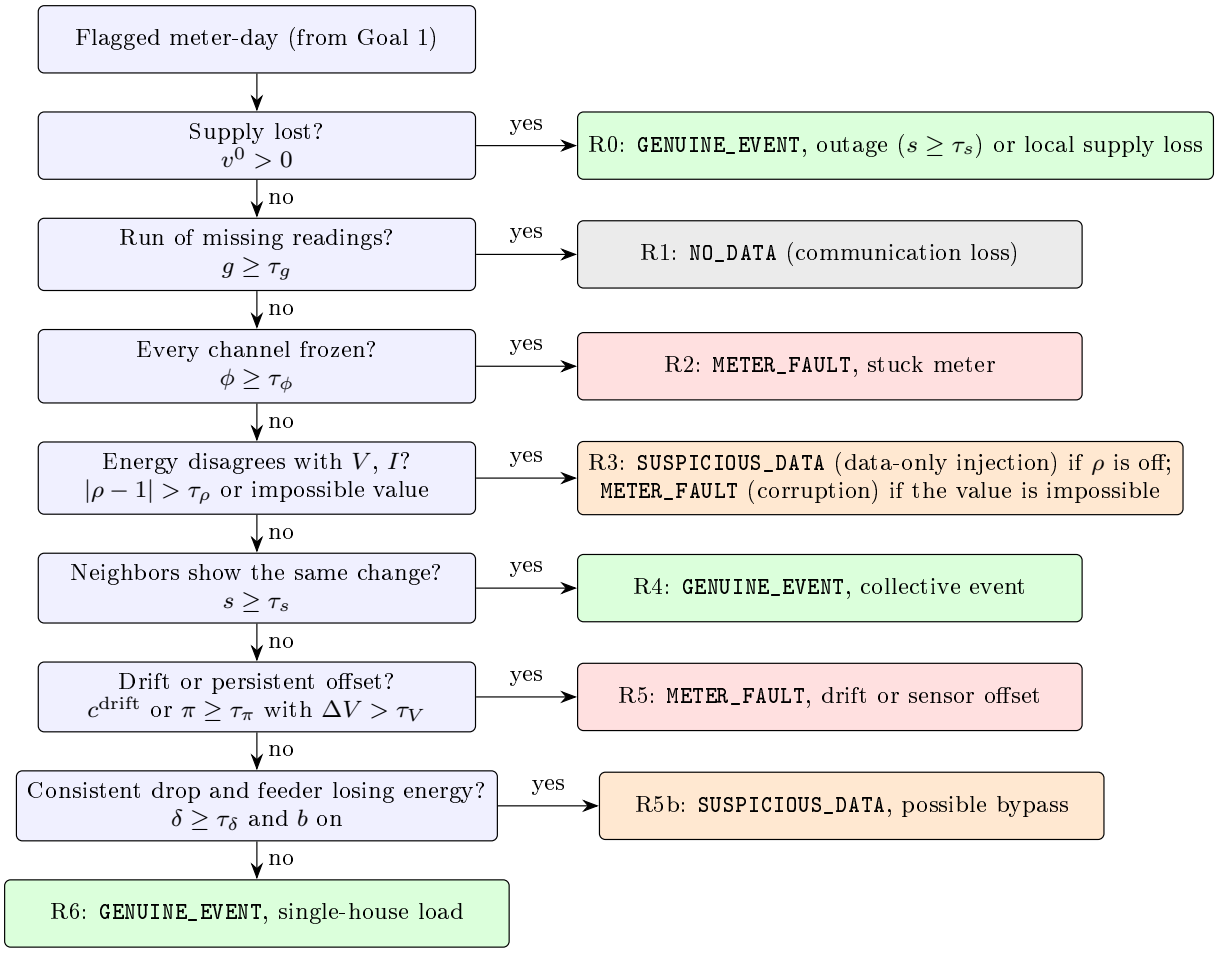


Figure 1: Rule cascade turning the evidence vector into a verdict. All thresholds τ are fitted by event-grouped cross-validation (Section 1.6) and stored in a versioned configuration file (`configs/thresholds.yaml`), not hard-coded.

Rule	Condition	Verdict	Reason sentence (template)
R0	$v^0 > 0$	`GENUINE_EVENT` (outage if $s \geq \tau_s$; local supply loss if 2–6 meters; single meter otherwise)	"Voltage was 0 V on {n} of {N} meters of feeder {f} for {hours} h: supply was lost and the meter is reporting correctly."
R1	$g \geq \tau_g$	`NO_DATA` (communication loss)	"{share} of this meter's readings did not arrive; no verdict is possible for this period."
R2	$\phi \geq \tau_\phi$	`METER_FAULT` (stuck meter)	"Energy and voltage have not changed for {hours} h; real voltage always varies."
R3	$\lvert\rho - 1\rvert > \tau_\rho$	`SUSPICIOUS_DATA` (data-only injection); `METER_FAULT` (corruption) if $E < 0$, $E = 99.999$ or $\times 1000$	"Reported energy is {ratio}× what voltage and current imply (limit {tau}); voltage and current look normal."
R4	$s \geq \tau_s$, ρ consistent	`GENUINE_EVENT` (collective)	"{n} of {N} meters on feeder {f} changed together and their electrical readings agree."
R5	$s < \tau_s$, c^{drift} on, or $\Delta V > \tau_V$ for $\geq \tau_\pi$ days	`METER_FAULT` (drift or sensor offset)	"This meter reads {dv} V above its feeder head, which is physically impossible, for {days} days."
R5b	$s < \tau_s$, ρ consistent, $\delta \geq \tau_\delta$, b on	`SUSPICIOUS_DATA` (possible bypass)	"This home reports {pct}% less than its usual level for {days} days while feeder {f} loses {gap}% more energy than usual; voltage and current agree with the reported energy, which fits a physical bypass."
R6	$s < \tau_s$, ρ consistent, short-lived	`GENUINE_EVENT` (single-house load)	"Only this home changed, briefly, and energy, voltage and current agree: an appliance switching on is the likely cause."

Table 3: Rule table. The tamper flag κ^{tamp} is reported as supporting evidence for R5b but never decides a verdict: in N1 it appears on 4 falsified readings and on 55 clean ones.

Notes on the rules.

- **R0 before R2.** Without it, every outage, load-shedding period, cable fault and transformer trip of N1 would be reported as a stuck meter, the opposite of the right answer.
- **R3 before R4.** A coordinated injection hits three meters of one feeder, so it could pass the neighborhood test; its energy disagrees with V and I, so R3 catches it first.
- **Suspect channel.** For R3 the suspect channel is energy by construction. For the other verdicts it is the feature with the largest autoencoder reconstruction error passed on by Goal 1; for faults that scale every channel (bias, bypass, frozen meter, clock shift) no single channel is named.
- **R5b before R6.** An isolated, electrically consistent drop is not automatically an appliance: when it lasts and the feeder starts losing energy at the same time, a bypass is the better

explanation. A short, isolated, consistent change with no balance alarm stays a single-house load.

- **What the rules cannot settle.** Goal 1 found that small sensor biases and gain drifts are rarely flagged at all, and that the balance alarm names a feeder, not a meter. When b is on but no meter shows a clear drop, the case is logged as `SUSPICIOUS_DATA` with a low confidence band, which sends it to a human (Goal 3).

1.6 Confidence Score

For each flagged meter-day the component outputs a probability for each of the four verdicts, $p_{i,t}(c)$ for $c \in \{\texttt{NORMAL}, \texttt{GENUINE_EVENT}, \texttt{METER_FAULT}, \texttt{SUSPICIOUS_DATA}\}$. The confidence of the reported verdict is $c_{i,t} = p_{i,t}(\hat{c})$, and the full vector is passed to Goal 3, which needs the margin between the two most likely verdicts and the set of plausible alternatives. The probabilities are built from three inputs, each of which can be inspected on its own:

1. the Goal 1 stack score $\tilde{a}_{i,t}$: the largest detector score, each standardized against its distribution on clean meter-days;
2. the *rule margin* $m_{i,t} \in [0, 1]$, which measures how clearly the rule that fired was satisfied. Each condition of that rule compares an evidence value e_j with a threshold τ_j. For each condition we compute its relative distance from the threshold, $\min\{1, |e_j - \tau_j|/\tau_j\}$, which is 0 when the value sits exactly on the threshold and 1 when it is at least one threshold-width away. The margin $m_{i,t}$ is the smallest of these values over the conditions of the rule, so a verdict is only as clear as its most borderline condition;
3. the *detector agreement* $\kappa_{i,t} \in \{0, \ldots, 6\}$: how many of the six Goal 1 detectors flagged the day, in the spirit of the explicit sigma rule used in rule-aware prompting [1].

Together with a one-hot code of the rule that fired, these inputs are mapped to probabilities by multinomial Platt scaling, one of the calibration techniques listed in the trust-calibration literature [3]: a multinomial logistic regression whose coefficients can be printed and read,

$$p_{i,t}(c) \;=\; \frac{\exp\big(\beta_{c,0} + \beta_{c,1}\tilde{a}_{i,t} + \beta_{c,2}m_{i,t} + \beta_{c,3}\kappa_{i,t} + \gamma_{c,r(i,t)}\big)}{\sum_{c'} \exp\big(\beta_{c',0} + \beta_{c',1}\tilde{a}_{i,t} + \beta_{c',2}m_{i,t} + \beta_{c',3}\kappa_{i,t} + \gamma_{c',r(i,t)}\big)}, \tag{1}$$

where $r(i,t)$ is the rule that fired. The target is the true class of the meter-day, known from the dataset labels. If the calibration curve is poor, isotonic regression, another technique from the same literature [3], is the alternative.

Confidence band. The probability is shown to operators as a band: HIGH above 0.90, LOW below 0.60, MEDIUM in between (Goal 3 thresholds). A class's band may be HIGH only if its calibration has been validated on at least 30 out-of-fold events. In N1 this holds for `GENUINE_EVENT` (37 events) and `METER_FAULT` (33), not for `SUSPICIOUS_DATA` (8 events, one per manipulation type), so `SUSPICIOUS_DATA` verdicts are capped at MEDIUM and are always reviewed by a human.

We do not assume that the result is calibrated; we measure it (Section 1.9). This matters because the confidence-based routing of Goal 3 assumes reasonably calibrated confidence: an overconfident model may delegate dangerous decisions to the AI, and an underconfident one may defer everything to humans [3].

Data used for fitting. A split by calendar period does not work on this dataset: in N1, November–December contain no falsification and no sensor-bias event, and September–October only two falsifications. Baselines (profiles, medians, scales) therefore use clean meter-days only,

and the thresholds τ and the coefficients β, γ are fitted by 5-fold cross-validation over the whole year, with folds grouped by event (`gt_event_id`), so that no event is split between fitting and testing; within each training fold, τ is fitted on one half of the events and β, γ on the other. Clean days are assigned to folds in meter-week blocks to keep near-duplicate neighboring readings from leaking between sets. All reported numbers are out-of-fold, per event, with Wilson 95% intervals.

1.7 From Evidence to Plain Language

Figure 2 shows how the verdict and the evidence become the text that the operator reads, following the chain rules and SHAP → evidence → structured summary → LLM text. There is no retrieval step (see "Why no retrieval" below). Full explanations are generated only for flagged meter-days, never for every normal reading, to keep latency and cost down.

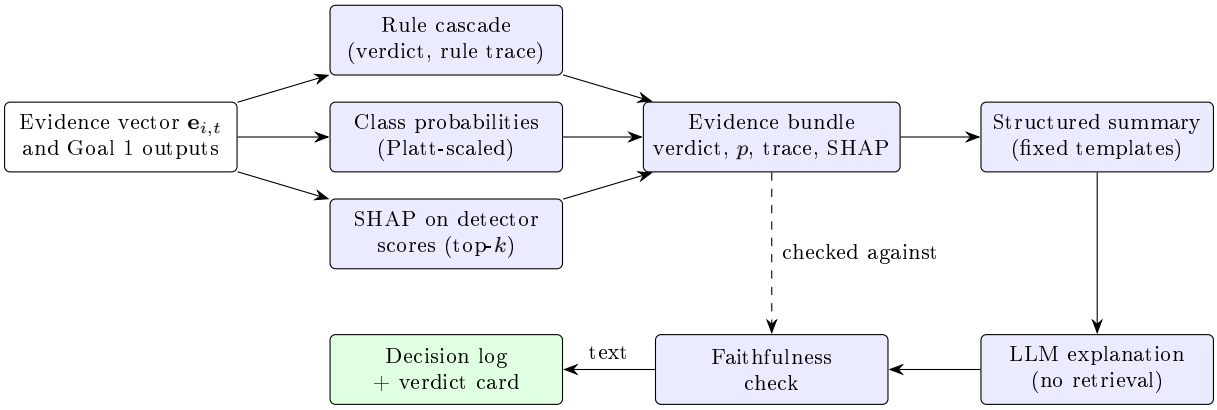


Figure 2: Explanation pipeline. The language model only rewrites the structured summary; it receives no retrieved documents, and its text is shown only if it passes the faithfulness check.

The language model receives only the evidence bundle and the structured summary, and writes about them; the faithfulness check compares its text with the same bundle. If the check rejects the text, the structured summary is logged and shown instead.

Step 1: attributions with SHAP

The rules say what kind of problem it is; SHAP says why the detectors flagged the meter-day in the first place. SHAP assigns each input feature a value showing how much it pushed a detector's score for this day up or down relative to a baseline, and it is model-agnostic. We apply the `shap` library to the Isolation Forest and the autoencoder of Goal 1, on their daily features, and only on flagged meter-days. Each feature is tied to a sentence template by a small lexicon, which is how a number becomes a statement (Table 4).

Goal 1 feature	Sentence template
Profile residual	"Today this home used {x} kWh against a usual {y} kWh for this season and day type."
Share of the feeder	"This home's share of feeder {f} has moved by {pct}% compared with the last four weeks."
Physics ratio	"Reported energy is {ratio}× what voltage and current imply."
V minus feeder-head V	"This meter reads {dv} V {above/below} the head of its feeder; usually {dv0} V."
Longest frozen run	"Energy and voltage stayed identical for {hours} h."
Lag gain against profile	"Today's pattern matches the usual one best when shifted by {h} h (clock problem?)."
Similarity to last week	"Today's readings are almost identical to the same day last week (possible replay)."

Table 4: Excerpt of the feature lexicon that turns attributions into sentences (one entry per Goal 1 feature; the evidence quantities of Table 2 have their own sentences in Table 3).

Step 2: structured evidence summary

A deterministic step assembles the evidence bundle into a summary: a headline (verdict, channel, confidence and band), the strongest reasons (the rule's reason sentence first, then the highest SHAP contributions), the runner-up verdict, a *what would change this verdict* line generated from the rule thresholds, and a suggested check. This follows the recommendation, in a review of AI approaches for smart-grid reliability, to combine local feature attribution, counterfactuals and logged explanations [5]. Every sentence comes from a template of Table 3 or Table 4, filled with values from the log. The summary is the input of the language model and also the text shown if the faithfulness check fails. An illustrative card (modelled on the data-only injection of feeder N1-F3 in May; values for illustration):

```
Event N1-2025-05-03-M026 | Meter N1-M026 (feeder N1-F3) | 2025-05-03
VERDICT: SUSPICIOUS DATA, energy channel     CONFIDENCE: 0.74 (MEDIUM)
Runner-up: METER FAULT (0.19)

Why:
 1. Reported energy is 0.65x what voltage and current imply
    (limit +/-5%) -> rule R3.
 2. Isolated: 0 of 9 other meters on feeder N1-F3 changed.
 3. Voltage and current follow this home's usual pattern; only the
    energy field is low.

This would be a METER FAULT instead if: the value were impossible
(negative, x1000, 99.999); a GENUINE EVENT if >= 3 neighbors changed
the same way and energy matched voltage and current.

Suggested check: read the meter register on site and compare it with
the reported energy; inspect the data path (gateway, head-end).
```

Step 3: LLM explanation with a faithfulness check

The language model turns the structured summary into a readable diagnosis in plain language. It receives the evidence bundle as structured data and nothing else. It is asked to justify the

verdict against an explicit, stated rule, as in rule-aware prompting for grid telemetry [1], and it may not alter the verdict, the confidence or the band.

Before the text is displayed, a faithfulness check compares it with the evidence bundle:

- every reason it names must be a fired rule or one of the top-k SHAP features;
- every number it quotes must match the logged value up to rounding;
- the verdict, the confidence and the band must be those in the log.

If any test fails, the structured summary is shown instead and the event is logged with that fact. We treat the LLM as a layer that describes the evidence, not as a source of reasoning.

Why no retrieval (RAG). We considered giving the language model retrieved context (descriptions of the fault kinds, the anomaly taxonomy, earlier log entries for the same meter). We dropped it. A study of retrieval-augmented LLM post-processing for streaming anomaly detection found only a modest gain, and the faithfulness of the explanations did not improve with retrieval [2]. For an operator who must verify a verdict, that small gain does not justify the extra latency on every flagged case and a document corpus to build and maintain. Everything the explanation needs is already in the evidence bundle.

1.8 Auditable Decision Log

Every flagged meter-day produces one record in an append-only JSON Lines file. Each record stores the hash of the previous one, so a later edit or deletion breaks the chain. The record is grouped around the three questions the challenge asks (Table 5); an abridged example follows. The escalation layer of Goal 3 appends its own `governance` block (route, rules fired, operator judgement, final decision) to the same record; this section defines only the fields produced by detection, conclusion and explanation. The full schema is `docs/decision_log_schema.json` in the project repository.

Block	Fields
`detected`	event id, day, node, feeder and meter id, model version, stack score, per-detector scores, per-feature reconstruction error, flag
`concluded`	verdict, sub-type, suspect channel, class probabilities, confidence, band
`why`	rule trace (rule id, measured value, threshold, passed or not), SHAP evidence, neighborhood support, feeder-balance alarm, threshold version
`explanation`	LLM diagnosis, result of the faithfulness check, text actually shown
`ground_truth`	filled afterwards, for evaluation only: dataset label, sub-type, event id

Table 5: Structure of a decision log record.


```
{"event_id":"N1-2025-05-03-M026","ts":"2025-05-03","node_id":"N1",
 "feeder_id":"N1-F3","meter_id":"N1-M026",
 "detected":{"model_version":"detect-0.1","fused_score":4.8,"flag":true,
   "detector_scores":{"rules":1,"robust_z":9.1,"svm":2.3,"ae":6.4}},
 "concluded":{"verdict":"SUSPICIOUS_DATA","subtype":"data_only_injection",
   "suspect_channel":"energy","confidence":0.74,"band":"MEDIUM",
   "class_probabilities":{"NORMAL":0.02,"GENUINE_EVENT":0.05,
                          "METER_FAULT":0.19,"SUSPICIOUS_DATA":0.74}},
 "why":{"rules":[{"id":"R3","measured":0.65,"threshold":0.05,"passed":true},
                 {"id":"R4","measured":0.0,"threshold":0.3,"passed":false}],
```

```
        "shap":[["physics_ratio",0.52],["profile_residual",0.21]],
        "neighbour_support":"0/9","balance_alarm":false,
        "thresholds_version":"0.1-phase1"},
 "explanation":{"llm_diagnosis":"...","faithfulness":"pass","shown":"llm"},
 "prev_hash":"9f2c...","hash":"41ab..."}
```


(The values are invented for illustration.) The record keeps the exact inputs and the version of the thresholds and models, so a verdict can be reproduced and reviewed later.

1.9 How We Will Evaluate Explanations and Confidence

The dataset labels every reading with its class, its exact sub-type and the event it belongs to, so verdicts and explanations can be tested against ground truth rather than only judged by eye. In N1 the flagged material comes from 98 events: 37 genuine events, 33 meter faults (25 of them data corruption), 8 falsifications and 20 communication losses. Most manipulation and sensor-fault types occur once, so results are reported per event, with intervals, and never as a single accuracy figure; we use precision, recall and false-positive rate per class.

Aspect	**Measure**	**Method**
Confidence quality	Calibration curve and Brier score [4]; verdict accuracy within each band	Out-of-fold, per class and overall
Verdict correctness	Precision, recall and false-positive rate per class, with event counts	Against the dataset labels, out-of-fold; also on the *subtle set*
Explanation correctness	Share of verdicts whose named channel is the true one (Table 7); share whose scope (isolated, part of the feeder, whole feeder) is right	Labels versus the `why` block
LLM faithfulness	Share of LLM texts accepted by the faithfulness check; kinds of statements rejected	Automatic check on all flagged meter-days
Does the LLM help	LLM text versus the plain structured summary, rated on clarity and correctness (1–5) by several team members who do not know which is which	Sample of flagged meter-days
Response latency	Median and 95th-percentile time of rules, SHAP and LLM per flagged meter-day	Timer on the evaluation run

Table 6: Evaluation plan for confidence scores and explanations.

True channel	Dataset sub-types
Energy	`fdi_scale`, `fdi_evasion`, `fdi_inflate`, `fdi_replay`, `coordinated_fdi`, all data corruption (`spike_gain`, `negative_value`, `saturation`, `unit_error_x1000`)
Voltage	`voltage_offset_drift`
All channels	`bias_step`, `gain_drift`, `stuck_value`, `clock_shift`, `partial_reduction_bypass`, `selective_bypass`, `meter_zeroing`: only the verdict and the scope are scored

Table 7: The dataset has no "injected channel" column; this fixed mapping from sub-type to channel is used to score the suspect channel named in the explanation.

The *subtle set* is built from existing events, without changing the dataset: the first seven days of the two gain drifts and of the bias step, and the evasion attack, whose scaling factor wanders to mimic natural variation.

1.10 Risks and Limitations

- **Few events per type.** Most manipulation and sensor-fault types occur once in N1, so per-type results are case studies, not rates; the `SUSPICIOUS_DATA` band is capped at MEDIUM for that reason.
- **Simulated data.** Thresholds and calibration coefficients are fitted on one generator's anomalies and may be optimistic; results on the subtle set are reported alongside.
- **Physically consistent manipulation.** Bypasses keep energy, voltage and current consistent; R5b depends on the feeder-balance alarm of Goal 1, which names a feeder and reacts after 1–2 weeks. A meter that under-reports slightly may never reach R5b.
- **PV homes.** The consistency check is skipped while a home exports, so an injection limited to export hours would be missed by R3.
- **Weak tamper log.** Tamper codes also appear on clean readings; they support but never decide a verdict.
- **Correlated features.** SHAP can spread credit among correlated features. The rule trace is therefore the primary explanation and SHAP the supporting one.
- **Cost of SHAP.** SHAP is run only on flagged meter-days and its latency is measured; the rule trace remains available if it is too slow.
- **Calibration can drift** across seasons or meters; the calibration curve is checked again on later events.
- **Convincing but wrong.** A fluent explanation can raise trust beyond what the evidence justifies, so every card shows the confidence band, the runner-up verdict and the counterfactual line.
- **Language model errors.** The faithfulness check and the fallback to the structured summary limit this, and the acceptance rate of the check is reported.

References

[1] "A Rule-Aware Prompt Framework for Structured Numeric Reasoning in Smart Grids," arXiv:2512.12794.

[2] "Component-Level Contributions of Retrieval-Augmented LLM Post-Processing in Streaming Anomaly Detection," *Sensors*, 2026.

[3] "Confidence-Based Trust Calibration in Human-AI Teams," SAI, vol. 16, no. 12.

[4] "Trust Calibration in AI," Emergent Mind overview page, https://www.emergentmind.com/topics/trust-calibration-in-ai.

[5] "Comprehensive review of AI-driven approaches for smart grid stability and reliability," ScienceDirect, 2025.

[6] "Time and Frequency Domain-based Anomaly Detection in Smart Meter Data," arXiv:2504.18231.

[7] "Multi-User Energy Consumption Monitoring and Anomaly Detection with Partial Context Information."